

ISE 5406: Optimization II

Lecture 15: Conjugate Gradient Methods

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Reading for this lecture

- Section 8.8 of Bazaraa et al. (2006)
- Chapter 5 of Nocedal and Wright

Conjugate Direction Method Recap

Convergence of CDM

Numerical Example

The Conjugate Gradient Method

Nonlinear CG Methods

Comparison of Methods

Exercises

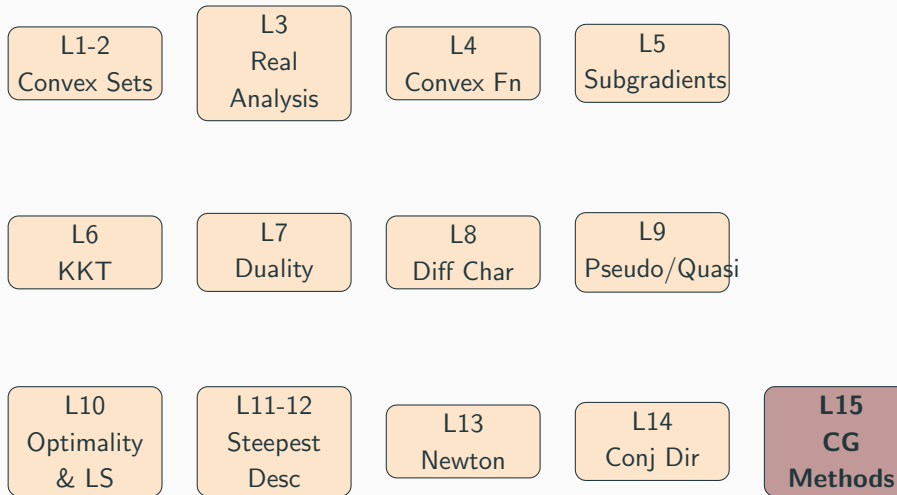
Last Lecture (L14): Conjugate Direction Method

- Definition of Q -conjugate directions
- Linear independence of conjugate directions
- Conjugate direction method (CDM) for quadratic functions
- Expanding subspace theorem: CDM converges in n steps

This Lecture (L15): Conjugate Gradient Methods

- CDM recap and convergence proof
- Numerical example of CDM
- Conjugate Gradient (CG) method: automatic direction generation
- CG algorithm and properties
- Nonlinear CG methods: Fletcher-Reeves and Polak-Ribière

Course Roadmap: Unconstrained Optimization



After L15, we transition to constrained optimization with equality

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CDM Setup: Convex Quadratic Programs

Consider the convex quadratic program:

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T Q \mathbf{x} + \mathbf{c}^T \mathbf{x},$$

where Q is positive definite.

Conjugate Direction Method:

- Given: starting point $\mathbf{x}_1 \in \mathbb{R}^n$ and n linearly independent Q -conjugate directions $\mathbf{d}_1, \dots, \mathbf{d}_n$.
- Iteration: $\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{d}_k$, where

$$\lambda_k = -\frac{\nabla_k^T \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k}$$

(obtained by minimizing $f(\mathbf{x}_k + \lambda \mathbf{d}_k)$ over λ)

- Here $\nabla_k = \nabla f(\mathbf{x}_k) = Q\mathbf{x}_k + \mathbf{c}$.

Key Property: At iteration k , we minimize f over the expanding

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Convergence Theorem

Theorem 2.1: Convergence in n Steps

For a convex quadratic function with positive definite Hessian, the conjugate direction method converges to the global minimizer in **at most n steps**, starting from any point $\mathbf{x}_1 \in \mathbb{R}^n$.

Proof Sketch:

1. Any point $\mathbf{x} \in \mathbb{R}^n$ can be expressed as:

$$\mathbf{x} = \mathbf{x}_1 + \sum_{i=1}^n \alpha_i \mathbf{d}_i$$

for some scalars α_i (since $\mathbf{d}_1, \dots, \mathbf{d}_n$ span $\mathbb{R}^n$).

2. The CDM step lengths $\boldsymbol{\lambda}^k = (\lambda_1, \dots, \lambda_k)$ satisfy:

$$\boldsymbol{\lambda}^k = \arg \min_{\boldsymbol{\alpha}^k \in \mathbb{R}^k} f(\mathbf{x}_1 + D^k \boldsymbol{\alpha}^k),$$

Proof: Expanding Subspace Property

Proof Details

Define $\theta_k(\alpha^k) := f(\mathbf{x}_1 + D^k \alpha^k)$. Expanding:

$$\begin{aligned}\theta_k(\alpha^k) &= \frac{1}{2}(\mathbf{x}_1 + D^k \alpha^k)^T Q(\mathbf{x}_1 + D^k \alpha^k) + \mathbf{c}^T(\mathbf{x}_1 + D^k \alpha^k) \\ &= \frac{1}{2}(\alpha^k)^T (D^k)^T Q D^k \alpha^k + (\mathbf{x}_1^T Q D^k + \mathbf{c}^T D^k) \alpha^k + f(\mathbf{x}_1).\end{aligned}$$

Since $Q \succ 0$ and D^k has full rank, $(D^k)^T Q D^k$ is positive definite, so θ_k is strictly convex.

The gradient with respect to α^k is:

$$\nabla_{\alpha^k} \theta_k = (D^k)^T Q D^k \alpha^k + (D^k)^T Q \mathbf{x}_1 + (D^k)^T \mathbf{c}.$$

At $\alpha^k = \lambda^k$, we have $\mathbf{x}_{k+1} = \mathbf{x}_1 + D^k \lambda^k$, so:

$$\nabla_{\alpha^k} \theta_k|_{\lambda^k} = (D^k)^T Q \mathbf{x}_{k+1} + (D^k)^T \mathbf{c} = (D^k)^T \nabla_{k+1} = 0^T.$$

The last equality holds because ∇_{k+1} is orthogonal to each $\mathbf{d}_i$ (by

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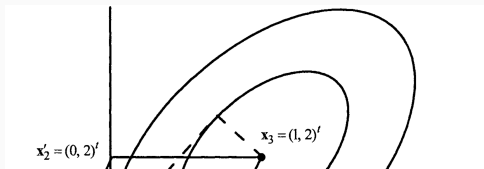
Exercises

Numerical Example: Example 8.8.2 (Bazaraa et al.)

Problem: Minimize $f(x_1, x_2) = 4x_1^2 + 4x_2^2 - 4x_1x_2 - 12x_2$ over $\mathbb{R}^2$.

Solution via CDM:

- Hessian: $Q = \begin{pmatrix} 8 & -4 \\ -4 & 8 \end{pmatrix}$ (positive definite)
- Starting point: $\mathbf{x}_1 = (0, 0)^T$, so $\nabla_1 = (0, -12)^T$
- First direction: $\mathbf{d}_1 = -\nabla_1 = (0, 12)^T$
- After 2 iterations: converges to optimal solution (as predicted by theory for $n = 2$)



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Motivation: Automatic Direction Generation

Challenge with CDM: How do we find n conjugate directions?

Idea: Generate conjugate directions iteratively using only gradient information.

Construction: At each iteration k , define the new direction as a linear combination of the negative gradient and the previous direction:

$$\mathbf{d}_{k+1} = -\nabla_{k+1} + \beta_k \mathbf{d}_k,$$

where β_k is chosen so that $\mathbf{d}_{k+1}$ is Q -conjugate to $\mathbf{d}_k$.

Computing β_k : Impose the conjugacy condition:

$$\begin{aligned}\mathbf{d}_{k+1}^T Q \mathbf{d}_k &= 0 \implies (-\nabla_{k+1} + \beta_k \mathbf{d}_k)^T Q \mathbf{d}_k = 0 \\ \implies \beta_k &= \frac{\nabla_{k+1}^T Q \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k}.\end{aligned}$$

Conjugate Gradient Algorithm

Initialize:

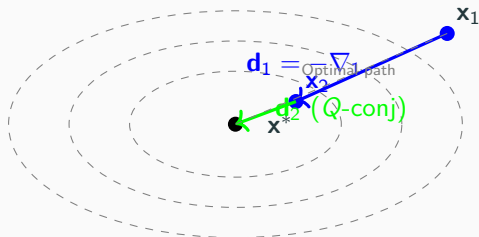
1. Choose starting point $\mathbf{x}_1 \in \mathbb{R}^n$
2. Set $\mathbf{d}_1 = -\nabla_1$ (where $\nabla_1 = \nabla f(\mathbf{x}_1) = Q\mathbf{x}_1 + \mathbf{c}$)

For $k = 1$ to n :

1. Compute step length: $\lambda_k = -\frac{\nabla_k^T \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k}$
2. Update iterate: $\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{d}_k$
3. Compute new gradient: $\nabla_{k+1} = Q\mathbf{x}_{k+1} + \mathbf{c}$
4. Compute conjugacy coefficient: $\beta_k = \frac{\nabla_{k+1}^T Q \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k}$
5. Update direction: $\mathbf{d}_{k+1} = -\nabla_{k+1} + \beta_k \mathbf{d}_k$

End

CG Method: TikZ Illustration on Elliptical Contours

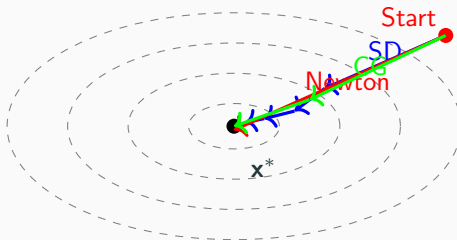


Left: Steepest descent direction $-\nabla_1$

Right: Conjugate direction $\mathbf{d}_2$ reaches $\mathbf{x}^*$ in 2 steps

For a 2D quadratic, CG converges in exactly 2 steps by automatically generating a conjugate direction!

Comparison: SD, Newton, and CG Convergence



Key Observations:

- **Steepest Descent (SD):** Oscillating, many steps (slow)
- **Newton:** Direct path (fast, but requires Hessian)
- **CG:** Smooth convergence, exploits problem structure (gradient-based, fast for n steps)

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Extension to Nonlinear Functions

Problem: CG requires the Hessian Q to compute β_k and λ_k , but:

- Hessian is expensive to compute
- Function may be nonquadratic

Idea 1: Taylor Approximation

For nonquadratic $f(\mathbf{x})$, use the second-order Taylor series at $\mathbf{x}_k$:

$$f(\mathbf{x}) \approx f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T (\mathbf{x} - \mathbf{x}_k) + \frac{1}{2} (\mathbf{x} - \mathbf{x}_k)^T \nabla^2 f(\mathbf{x}_k) (\mathbf{x} - \mathbf{x}_k).$$

This is locally quadratic, so CG can be applied.

Idea 2: Approximate β_k Without Hessian

Replace $\beta_k = \frac{\nabla_{k+1}^T Q \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k}$ with approximations using only gradients:

1. **Fletcher-Reeves (FR):** $\beta^{\text{FR}} = \frac{\nabla_{k+1}^T \nabla_{k+1}}{\nabla_k^T \nabla_k}$

Nonlinear CG: Algorithm and Intuition

Nonlinear CG Algorithm

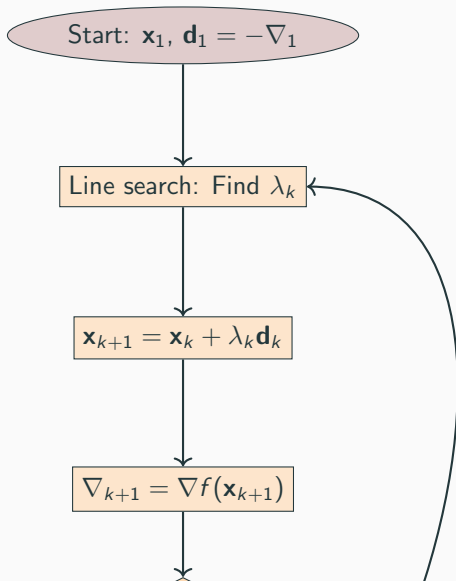
Initialize:

1. Choose starting point $\mathbf{x}_1$
2. Set $\mathbf{d}_1 = -\nabla_1$ (steepest descent direction)

For $k = 1, 2, \dots$:

1. **Line search:** Find λ_k minimizing $f(\mathbf{x}_k + \lambda \mathbf{d}_k)$
(e.g., via Wolfe conditions, not exact minimization)
2. Update: $\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{d}_k$
3. Compute: $\nabla_{k+1} = \nabla f(\mathbf{x}_{k+1})$
4. Check convergence: Stop if $\|\nabla_{k+1}\| < \epsilon$
5. Compute: β_k (using FR or PR formula)
6. Update: $\mathbf{d}_{k+1} = -\nabla_{k+1} + \beta_k \mathbf{d}_k$

CG Algorithm Flowchart



Roadmap

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Comparison: SD vs. Newton vs. CG

Property	Steepest Descent	Newton	CG
Convergence (quadratic)	$O(\kappa^2)$	Quadratic	Finite (n steps)
Hessian required?	No	Yes	No (for CG)
Computation per step	Low	High	Low–Medium
Memory	$O(n)$	$O(n^2)$	$O(n)$
Suitable for large n ?	Moderate	No	Yes
Extends to nonquadratic?	Yes	Yes	Yes (FR/PR)
Condition number sensitive?	Yes (κ^2)	No	Moderate

Where $\kappa = \lambda_{\max}(Q)/\lambda_{\min}(Q)$ is the condition number.

For high-dimensional problems or ill-conditioned systems, CG is usually the best choice among these three.

When to Use Each Method

Use Steepest Descent if:

- Problem dimension is small ($n < 100$)
- Function is highly nonlinear (gradients only available)
- You need a simple, robust baseline method

Use Newton's Method if:

- Second derivatives are readily available
- Quadratic convergence is critical
- Problem dimension is moderate ($n < 1000$)

Use Conjugate Gradient if:

- Problem dimension is large ($n > 1000$)

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Exercise 1: CDM Convergence

Quick Check: CDM Convergence

True or False?

1. The conjugate direction method converges to the optimum in exactly n steps for any convex quadratic problem. - - -
2. If the directions $\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3$ are Q -conjugate, then the CDM will compute the same step lengths $\lambda_1, \lambda_2, \lambda_3$ regardless of the order in which these directions are used. - - -
3. The expanding subspace theorem says that after k steps of CDM, we have minimized f over an $(k + 1)$ -dimensional subspace. - - -

Exercise 2: CG by Hand on 2x2 Problem

CG Step Computation

Consider $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T Q \mathbf{x} - \mathbf{c}^T \mathbf{x}$ with $Q = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

Starting from $\mathbf{x}_1 = (1, 1)^T$:

1. Compute $\nabla_1 = \nabla f(\mathbf{x}_1)$. (Hint: $\nabla f(\mathbf{x}) = Q\mathbf{x} - \mathbf{c}$)
2. Set $\mathbf{d}_1 = -\nabla_1$ and compute $\lambda_1 = -\frac{\nabla_1^T \mathbf{d}_1}{\mathbf{d}_1^T Q \mathbf{d}_1}$.
3. Update: $\mathbf{x}_2 = \mathbf{x}_1 + \lambda_1 \mathbf{d}_1$.
4. Compute ∇_2 and verify that the method converges in 2 steps (by checking ∇_2 is negligible or orthogonal to $\mathbf{d}_1$).

Exercise 3: Q-Conjugacy Verification

Verify Q-Conjugacy of CG Directions

The CG method generates directions via $\mathbf{d}_{k+1} = -\nabla_{k+1} + \beta_k \mathbf{d}_k$ with $\beta_k = \frac{\nabla_{k+1}^T Q \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k}$.

Task: Show that $\mathbf{d}_{k+1}^T Q \mathbf{d}_k = 0$ (i.e., $\mathbf{d}_{k+1}$ and $\mathbf{d}_k$ are Q-conjugate).

$$\begin{aligned}\mathbf{d}_{k+1}^T Q \mathbf{d}_k &= (-\nabla_{k+1} + \beta_k \mathbf{d}_k)^T Q \mathbf{d}_k \\&= -\nabla_{k+1}^T Q \mathbf{d}_k + \beta_k \mathbf{d}_k^T Q \mathbf{d}_k \\&= -\nabla_{k+1}^T Q \mathbf{d}_k + \frac{\nabla_{k+1}^T Q \mathbf{d}_k}{\mathbf{d}_k^T Q \mathbf{d}_k} \cdot \mathbf{d}_k^T Q \mathbf{d}_k \\&= \boxed{?}\end{aligned}$$

Exercise 4: Fletcher-Reeves vs. Polak-Ribière

Comparing Nonlinear CG Variants

For the nonlinear function $f(\mathbf{x}) = x_1^2 + 4x_2^2 + 2x_1x_2 - 2x_1 - 4x_2$:

Starting from $\mathbf{x}_1 = (0, 0)^T$, compute the first step of:

1. **Fletcher-Reeves (FR):** $\beta_1^{\text{FR}} = \frac{\nabla_2^T \nabla_2}{\nabla_1^T \nabla_1}$
2. **Polak-Ribière (PR):** $\beta_1^{\text{PR}} = \frac{\nabla_2^T (\nabla_2 - \nabla_1)}{\nabla_1^T \nabla_1}$

Which gives a larger β_1 ? How does this affect the search direction $\mathbf{d}_2$?

Exercise 5: Method Selection Scenario

When to Use Which Method?

For each scenario below, recommend whether to use **Steepest Descent (SD)**, **Newton**, or **Conjugate Gradient (CG)**.

1. Training a neural network with 1 million parameters; gradient computation is fast, but second derivatives are prohibitive.

2. Minimizing a small 50-parameter quadratic function where the Hessian is dense and easily invertible.

3. Solving a convex problem with 10,000 parameters; sparse Hessian structure is available, but Hessian-inverse is expensive.

Exercise 6: Convergence Rate Understanding

Convergence Rate Comparison

Consider a quadratic problem with condition number $\kappa = 100$.

1. For Steepest Descent, the error at step k is roughly $O(\kappa^{-2k})$. How many steps are needed to reduce the error by a factor of 10^6 ?

----- steps

2. For Conjugate Gradient, convergence is guaranteed in n steps (or fewer due to clustering of eigenvalues). If $n = 100$, what is the maximum number of steps?

----- steps

3. Which method is faster for this problem, and why?

Exercise 7: End-of-Lecture Assessment

True or False Statements

1. The Conjugate Gradient method requires storing all previous search directions to generate a new direction.
- - -
2. For a general nonquadratic function, the CG method with line search is guaranteed to converge in n steps.
- - -
3. The Fletcher-Reeves formula for β_k uses only gradient information and does not require the Hessian. - - -
4. Steepest Descent always converges faster than Conjugate Gradient on ill-conditioned quadratic problems. - - -
5. The expanding subspace theorem implies that after k steps of CDM, the iterate $\mathbf{x}_{k+1}$ is the global minimizer

Summary: Unconstrained Optimization Arc (L10–L15)

Key developments across the last 6 lectures:

- **L10:** Optimality conditions and line search foundations
- **L11–12:** Steepest Descent — simple, general-purpose, but slow on ill-conditioned problems
- **L13:** Newton's Method — fast quadratic convergence, but requires (and stores) Hessian
- **L14:** Conjugate Direction Method — exploits problem structure via conjugacy
- **L15 (Today):** Conjugate Gradient — automatic direction generation, scales to large n

Takeaways:

- **CDM Convergence:** Any convex quadratic is solved in $\leq n$ steps

Next Lecture (L16): Constrained Optimization with Equality Constraints

Transition: From unconstrained to constrained optimization.

Next Lecture Topics:

- Equality constraints and feasible regions
- Lagrange multipliers: geometric and algebraic interpretation
- KKT conditions for equality-constrained problems
- Lagrangian function and saddle point characterization
- Methods for equality-constrained optimization (e.g., null space method, penalty methods)

Why this matters:

- Many real-world problems have hard constraints (e.g., conservation